

PROBLEM & MOTIVATION

Solar energy has emerged as one of the most promising renewable energy sources for addressing global climate change and reducing dependence on fossil fuels. As photovoltaic (PV) technology becomes increasingly cost-effective and widespread, the integration of solar power into electrical grids presents both opportunities and challenges. The inherent variability of solar power generation due to changing meteorological conditions creates significant difficulties for grid operators who must continuously balance energy supply and demand while maintaining grid stability and reliability. Traditional forecasting methods often face significant limitations in handling non-linear relationships between multiple meteorological parameters and solar output. Machine learning approaches offer advantages by effectively capturing complex patterns without requiring explicit mathematical formulation of physical relationships. This project develops an advanced machine learning system for solar energy output prediction using comprehensive weather data analysis from the Israel Meteorological Service (IMS). The system implements a dual-architecture approach combining Gated Recurrent Unit (GRU) networks for short-term forecasting (24 hours) and Autoformer transformer variants for long-term prediction (up to 96 hours), addressing the fundamental challenge that different forecasting horizons require specialized architectures.

TECHNOLOGY STACK

- Machine Learning

 - TensorFlow, Keras, PyTorch
 - Python, NumPy, Pandas
- Frontend

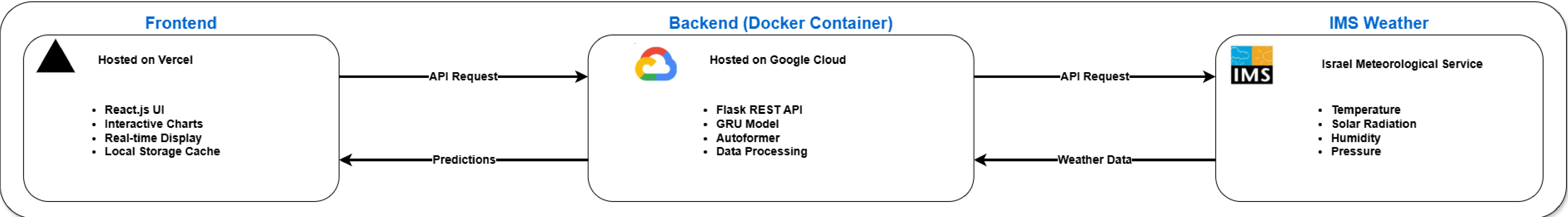
 - React.js
 - Chart.js
 - Vercel Hosting
- Backend

 - Flask REST API
 - Python
 - Docker Container
 - Google Cloud Platform
- Data Source

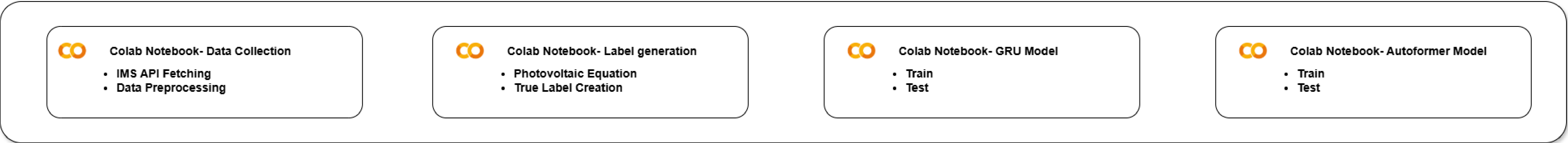
 - Israel Meteorological Service API

SYSTEM ARCHITECTURE

PRODUCTION SYSTEM

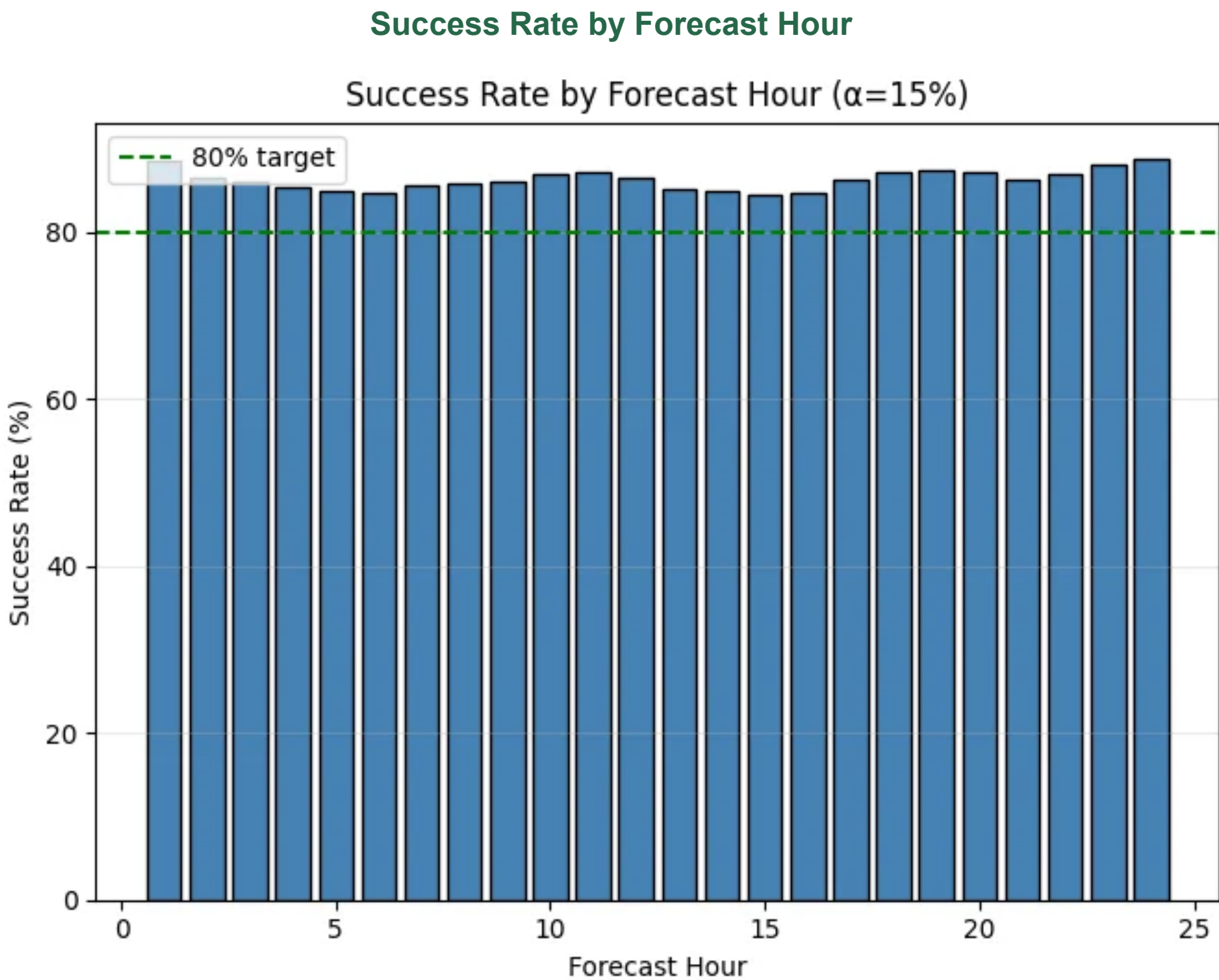
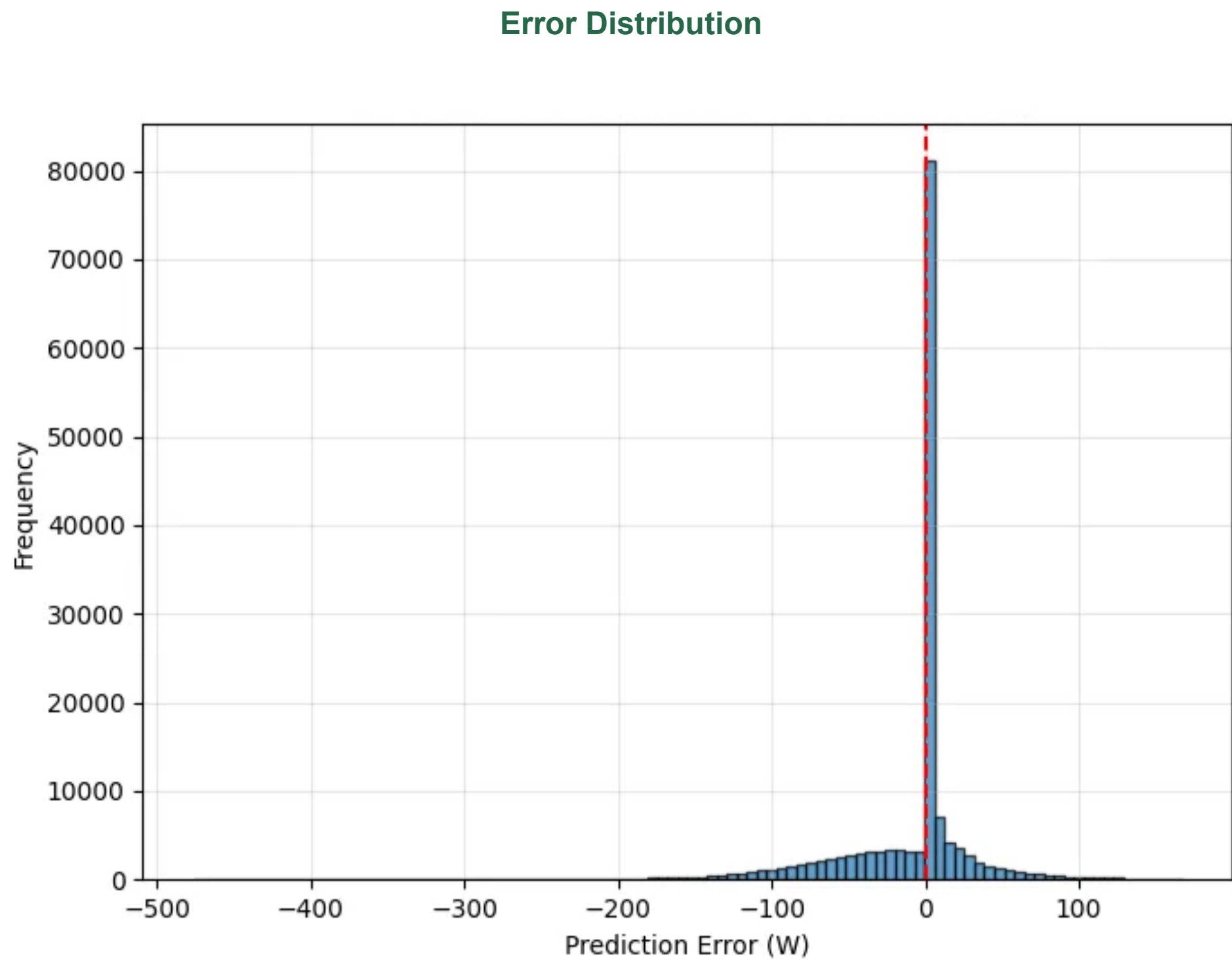
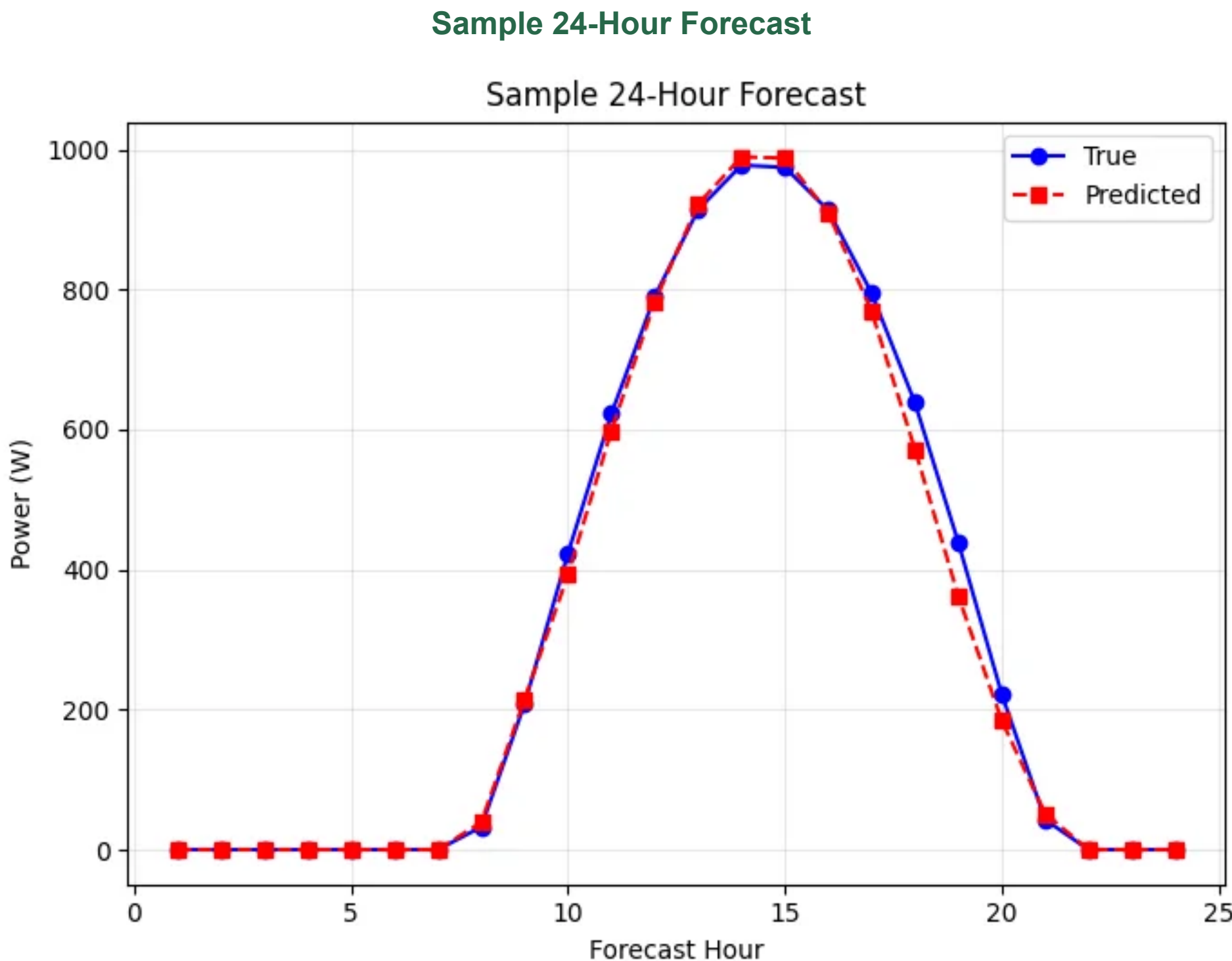


Development Utils



GRU MODEL - SHORT-TERM FORECASTING (24 HOURS)

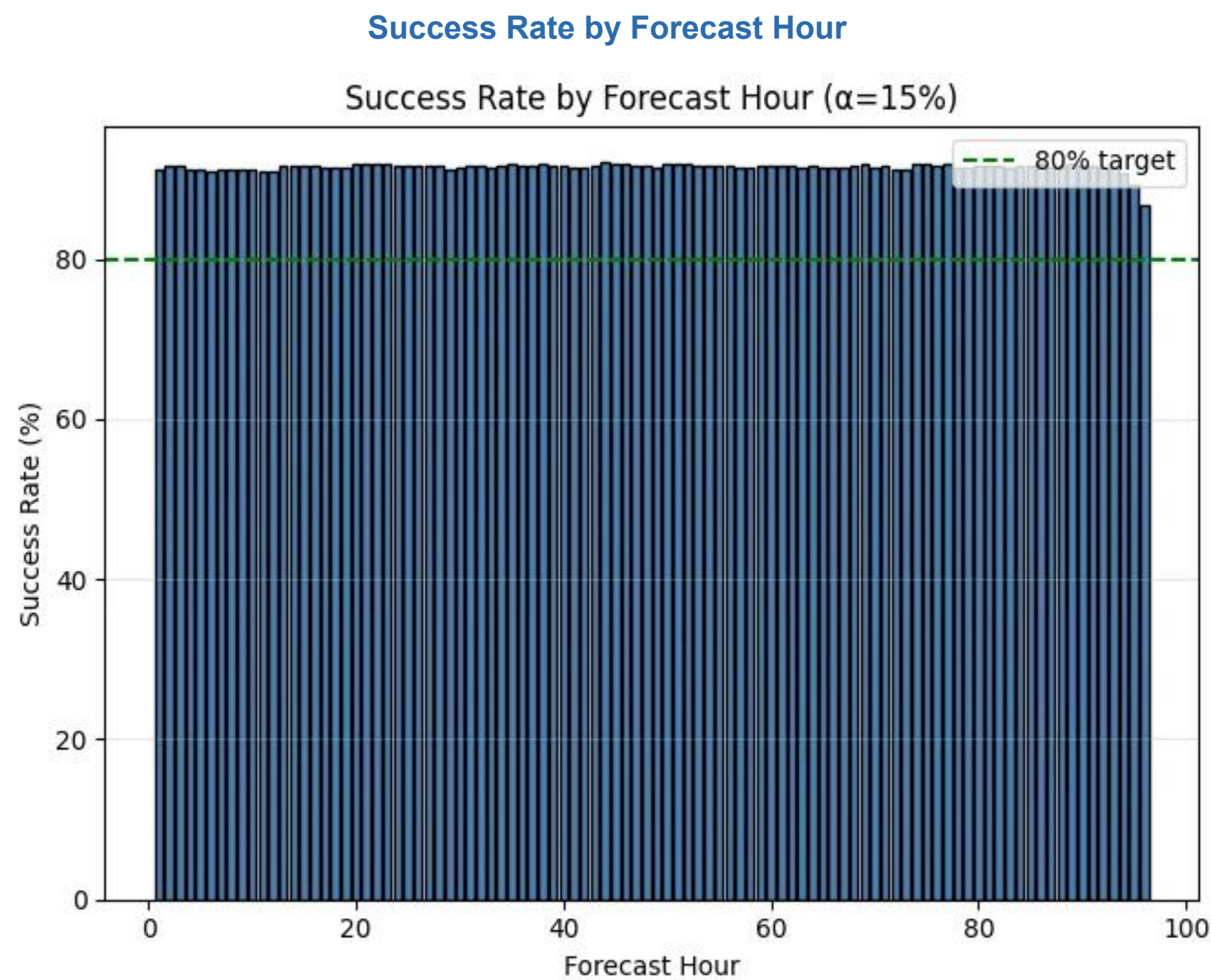
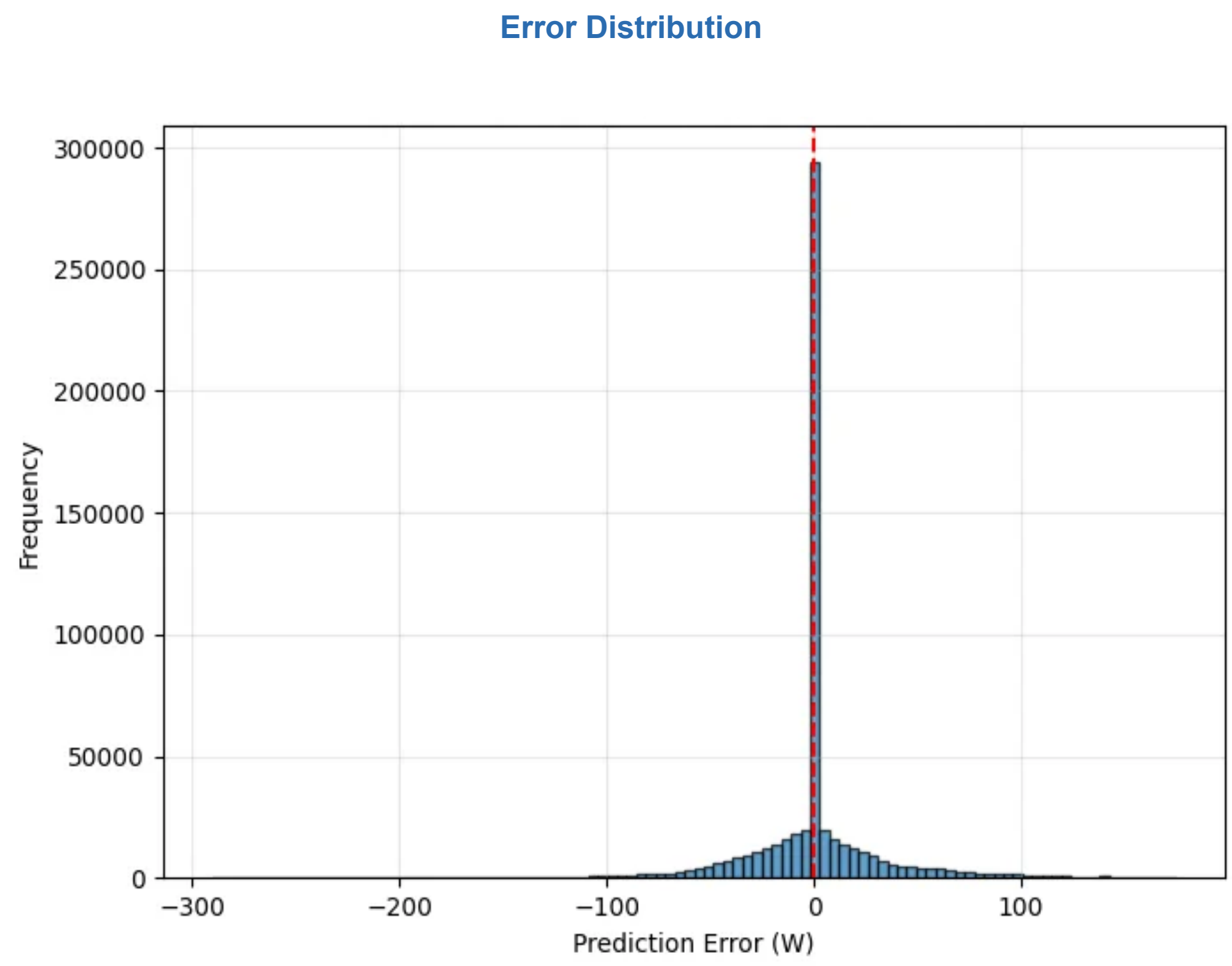
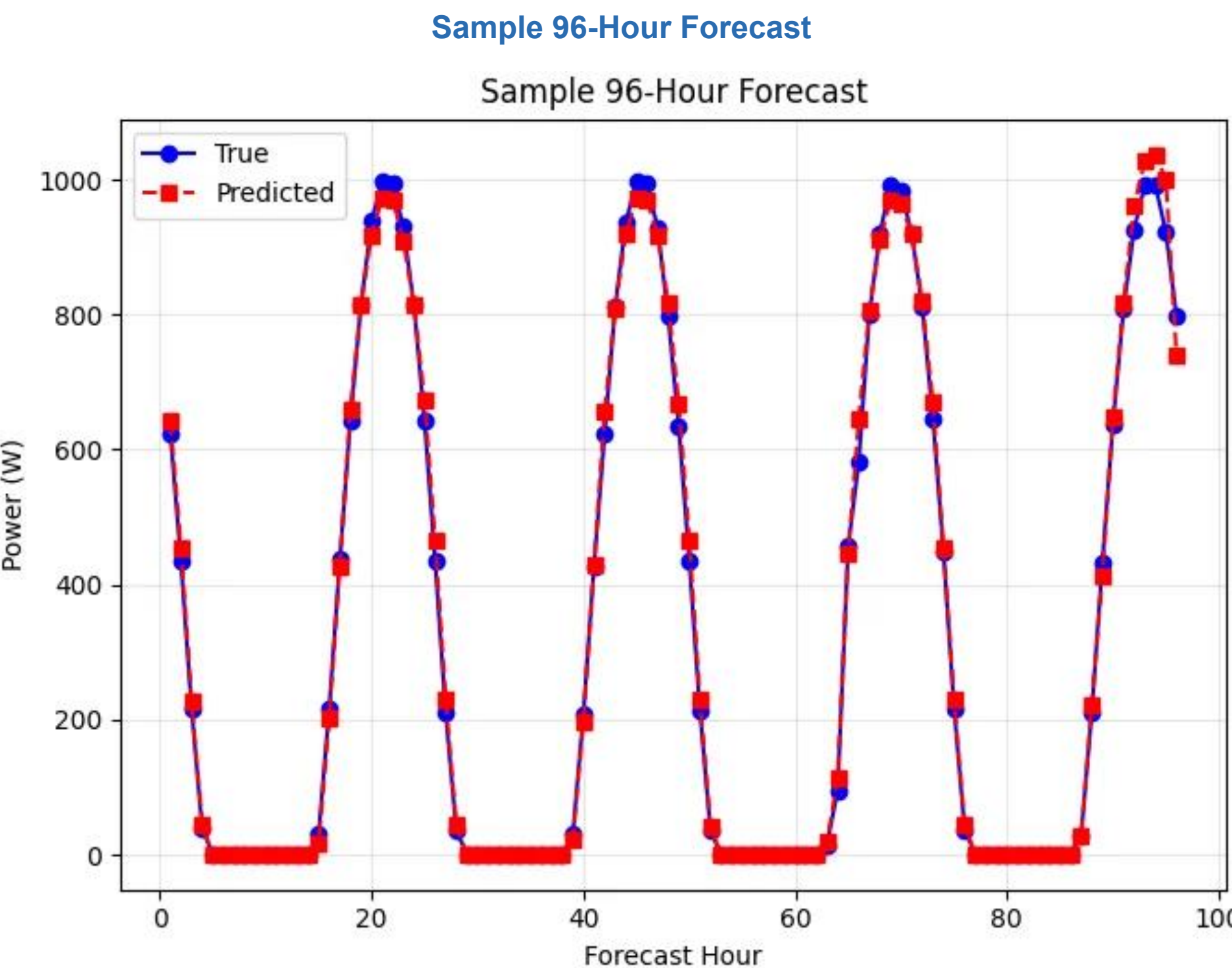
Gated Recurrent Unit (GRU) networks provide sequential hourly predictions using update and reset gates for temporal pattern learning. The model processes 72 hours of historical weather data to generate next-day forecasts with high accuracy for operational grid management.



Success Rate: 86.32% ($\alpha=15\%$) | Mean Absolute Error of Misses: 86.83

AUTOFORMER MODEL - LONG-TERM FORECASTING (96 HOURS)

Autoformer is a specialized transformer variant using autocorrelation mechanisms to capture seasonal and trend patterns. The encoder-decoder architecture with series decomposition enables direct multi-step forecasting for strategic energy planning. The model uses 336 hours (two weeks) of historical weather data to generate 4-day forecasts.



Success Rate: 91.5% ($\alpha=15\%$) | Mean Absolute Error of Misses: 68.24

CONCLUSIONS

The dual-architecture approach successfully addresses multi-horizon forecasting challenges. GRU networks provide computational efficiency for short-term operational needs, while Autoformer's autocorrelation mechanisms excel at capturing seasonal patterns for strategic planning.

Both models exceed the 80% success rate target ($\alpha=15\%$), demonstrating reliable performance for grid integration and energy planning applications.

KEY REQUIREMENTS

- ✓ Success Rate $\geq 80\%$ for forecasts
- ✓ Multi-day predictions (up to 4 days)
- ✓ hourly weather data integration
- ✓ Prediction response time ≤ 30 seconds
- ✓ Web-based interactive dashboard

CHALLENGES OVERCOME

- Military reserve duty during wartime
- No access to real solar farm data
- Theoretical PV calculation adoption
- Compressed development timeline
- Work-study balance constraints